

Psychology and Neuroscience 43rd Annual Graham Goddard In-House Conference



May 4 and 5, 2017
Life Sciences Centre
Room 5260

The 2017 Psychology and Neuroscience Graham Goddard In-House Conference

Welcome! We would like to thank you for attending the 43rd Annual Graham Goddard In-House Conference. This is an excellent opportunity to hear about the ongoing research of students, faculty and colleagues. It is your participation that makes this event a success.

Thanks should also be extended to the In-House Conference Committee. The time that the following individuals put into the planning and organization of the In-House is greatly appreciated: Kevin Duffy, R.J. Redden, Kevin LeBlanc, Francesco Usai, Chris Friesen, Kiera O'Neil, Lynn MacKenzie, Emre Fertan, Susan Lowerison, and Suzanne King.

Dr. Graham V. Goddard

Dr. Graham V. Goddard (1938-1987) was a leading researcher at Dalhousie University. Goddard invented the kindling model of epilepsy that has, since its introduction in the 1960s, become widely used research tool in studies of long-term potentiation (LTP) and epilepsy. His work was published in Nature, Science, along with other high profile journals. During his time at Dalhousie, Goddard helped establish Dalhousie as a leading research institution. Goddard also organized the annual Psychology Department In-House Conference, to help investigators in the department update others about their work and encourage academic dialogue. It is in his honour that we continue this tradition today.

Friendly Reminders

Talks will be held in LSC 5260.

Coffee and tea will be available in the 4th Floor Study Centre (4316).

Lunch on Friday, May 5th will take place in the 2nd Floor Lounge.

There will be pizza and drinks to follow at the end of Friday, May 5th, in the 2nd Floor Lounge.

Thursday, May 4, 2017 Room 5260

1:30 – 1:45 Introduction by Dr. Tara Perrot

1:45—3:00 Session 1 Chair: Dr. Aaron Newman

- 1:45 **T1) First-time mothers' attributions for postpartum sexual changes: Associations with sexual and relationship well-being**
Vannier SA, Adare K, Rosen NO
- 2:00 **T2) Understanding the Neural Bases for Social Deficits in Alzheimer's Disease**
Kosel, F, Franklin T
- 2:15 **T3) RadarWatch: Validating an attention training game with a focus on vigilance decrement**
Patrick E, Drake R, Kroeker J, Chisholm J, Klein R, Westwood D, Eskes G
- 2:30 **T4) God and Karma: Exploring Generosity Levels among Christians and Buddhists in Hong Kong**
Curtis R
- 2:45 **T5) Item-based Rasch Analysis of the Merrill-Palmer-Revised Scales of Development for Children with Autism Spectrum Disorder**
Dempsey EE, Smith IM
- 3:00 **T6) When *white* is red in the eye of the *Drosophila* beholder, and philosophy 101**
Shaw S

3:15—3:45 Coffee Break (LSC 4316—Study Centre)

3:45—5:00 Session 2 Chair: Dr. Ray Klein

- 3:45 **T7) FoxP2 Protein Expression in Zebra Finches Reared by Single Parents**
Gayda E
- 4:00 **T8) Cue-induced Functional Connectivity in the Nucleus Accumbens and Anterior Cingulate Gyrus in Smokers with and without Psychosis**
Ellsworth-Spotton G
- 4:15 **T9) Employing Rescorla's truly random condition to measure truly exogenous covert attention**
Habibnezhad M, Lawrence M, Klein R
- 4:30 **T10) Pediatric Insomnia Treatment in the Context of Obstructive Sleep Apnea: A Usability and Pilot Study**
Isaacs J, Corkum P, Godbout R, Witmans M
- 4:45 **T11) The influence of Pavlov on Lashley and Hebb**
Brown RE

Friday, May 5, 2017 Room 5260

9:00—10:30 Session 3 Chair: Dr. Tamara Franklin

- 9:00 **T12) The impact of fetal and neonatal experience on mammalian brain development and stress responsivity**
Lee K, Weaver I, Goguen D
- 9:15 **T13) Emotion Regulation in Couples Affected by Female Sexual Interest and Arousal Disorder**
Dubé J, Rosen NO
- 9:30 **T14) The Moderating Role of Self-Compassion on the Relation between Alcohol-Involved Sexual Assault and Negative Emotional Outcomes in Undergraduate Men and Women**
Strickland NJ, Stewart SH, Kehayes I-L, Wekerle C, Moore C, Caring Campus Team
- 9:45 **T15) Behavioural, Molecular and Epigenetic consequences of Perinatal Infection throughout Development and into Adulthood.**
Wheeler RV, Alam S, Weaver ICG, Franklin TB
- 10:00 **T16) Is there a relationship between working memory and reading in university students?**
Rehan S, Rubinfeld L, Eskes G
- 10:15 **T17) Male Experience, Female Investment and Offspring Anxiety Behavior: Where Father's Nature Meets Mother's Nurture**
Korgan A

10:30—11:00 Coffee Break (LSC 4316—Study Centre)

11:00—12:15 Session 4 Chair: Dr. Shelley Adamo

- 11:00 **T18) Perfectionistic Concerns Confer Risk for Anxiety Symptoms: A Meta-Analysis of 11 Longitudinal Studies**
Vidovic V, Smith MM, Sherry SB, Stewart SH, Saklofske DH
- 11:15 **T19) Examining Links Between Sexual Savouring and Sexual and Relationship Well-Being**
Richardson HM, Vannier SA, Rosen NO
- 11:30 **T20) Self-Distraction During Exercise in Clients with High (versus Low) Anxiety Sensitivity**
Pridy C, Watt M, Stewart SH
- 11:45 **T21) Investigating the neural signature of Inhibition of Return using a multimodal paradigm**
MacNeil H, Klein R, d'Entremont G
- 12:00 **T22) On the choice of animal models in neuroscience**
Mitchell DE

12:15—1:45 Lunch and Poster Session 2nd Floor Lounge

- P1) Investigation of neuron structural modifications within the visually deprived superior colliculus.**
NESC3370: Zajdlik K, Lausch S, Crouse C, Elia T, Aronitz E, Ellis S, Fish J, Lagman P, MacDonald J, McGroarty E, Mustafa D, Piercey M, Bassil S, Connell L, Lee S, MacLellan K, Pappin J, Ribeiro G, Krautner S, Duffy KR
- P2) A Brain Museum Tour of Europe**
Brown RE
- P3) Does active sharing lead to increased positive affect in children?**
Puddicombe E, Pratt T, Longard J, Moore C
- P4) eHealth intervention recruitment: Relative success of digital, media, and traditional recruitment methods for Better Nights, Better Days**
Rigney G, Chambers CT, Godbout R, Gruber R, Hall W, Reid G, Stremmler R, Weiss SK, Witmans M, Begum EA, Corkum P
- P5) On the Time-Course of Cued Temporal Attention**
McCormick C, Redden R, Lawrence M, Klein R
- P6) Novice Inter-Rater Reliability in Spike Sorting: Tetrodes vs. Single Channel Electrodes**
Shapiro JT, Korgan AC, King JL, Crowder NA
- P7) Assessing Attention: A tablet-based proof of concept with the Dalhousie Computerized Attention Battery (DalCAB)**
Patrick E, Cutmore L, Kroeker J, Jones S, Salmon J, Eskes G
- P8) 15 Years of ANT Research: Exploring the evolution of the Attention Network Test**
Arora S, Klein R
- P9) Utilization of Research Electronic Data Capture (REDCap) in a web-based Randomized Controlled Trial**
Begum EA, Corkum P
- P10) Major connections in glomerulus VA1v in the *Drosophila* antennal lobe**
Horne, JA, McLin S, Wiederman M, Borycz JA, Tarnogorska D, FlyEM Project Team, Plaza S, Xu S, Hess H, Meinertzhagen IA

1:45—3:00 Session 5 Chair: Dr. Ian Weaver

- 1:45 **T23) The roles of parvalbumin and somatostatin expressing interneurons in modulating contrast adaptation in mouse primary visual cortex**
King JL, Korgan AC, Crowder NA
- 2:00 **T24) The Effect of French Immersion on Native English Processing in the Brain**
Tagarelli KM, Lazar-Kurz Z, Newman AJ
- 2:15 **T25) Exploring Parent-Reported Perioperative Distress Behaviours in Children with Autism Spectrum Disorder**
Cameron E, Snow S, Smith I, Chorney J
- 2:30 **T26) Potential molecular conflict between detoxification and immune defense in *Manduca sexta***
Hall LE, Adamo SA
- 2:45 **T27) Illness-induced anorexia may help promote immune function by reducing the use of glutathione, a limited resource, by detoxification pathways.**
Miller D, Hall L, Adamo S

3:00—3:30 Coffee Break (LSC 4316—Study Centre)

3:30—4:45 Session 6 Chair: Dr. Leslie Phillmore

- 3:30 **T28) Susceptibility to monocular deprivation following immersion in darkness just prior to the critical period peak**
Lingley A, Duffy KR
- 3:45 **T29) Drinking frequency, drinking quantity, and alcohol-related problems as trait-states: A 4-wave, 18-month longitudinal study of female undergraduates**
Bartel, SJ, Sherry SB, Nealis LJ, Collins JL, Lee-Baggley DL, Stewart SH
- 4:00 **T30) Orthographic knowledge as a predictor of gains in lexical access**
MacKay E, Deacon H, Conrad N
- 4:15 **T31) The Connectome of a *Ciona intestinalis* larva**
Ryan K
- 4:30 **T32) A second complete connectome: the larval CNS of the ascidian *Ciona intestinalis***
Meinertzhagen IA, Lu Z, Ryan K

5:00-7:00 Dinner and Reception 2nd Floor Lounge

T1) First-time mothers' attributions for postpartum sexual changes: Associations with sexual and relationship well-being

Vannier SA, Adare K, Rosen NO

The transition to parenthood presents women with notable sexual and relationship challenges. There is considerable research examining physiological predictors of postpartum sexual and relationship well-being, but less is known about psychological factors. *Causal attributions* for sexual changes may be of particular importance because one specific cause may be difficult to pinpoint given the wide range of postpartum sexual concerns and problems among first-time parents. Attributions can be conceptualized on four dimensions: internal/external (i.e., is the event due to the self or the situation), stable/unstable (i.e., will the event occur again in the future), global/specific (i.e. will the cause of the event affect other situations), and partner responsibility (i.e., is the partner the cause). In the current study 117 first-time mothers completed measures assessing attributions for postpartum sexual changes, sexual satisfaction, functioning, and distress, and relationship satisfaction. Internal, stable, and global attributions were associated with greater sexual distress and lower relationship satisfaction. Partner attributions were associated with lower sexual and relationship satisfaction. These associations remained after controlling for time postpartum, breastfeeding, sexual frequency, and relationship duration. As postpartum sexual changes are common, attributions for these changes may be a valuable target for intervention by clinicians and healthcare professionals.

T2) Understanding the Neural Bases for Social Deficits in Alzheimer's Disease

Kosel, F, Franklin T

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder characterized by abnormal accumulation of β -amyloid protein in the brain. Due to the widespread nature of the disease, lesions can be seen throughout the brain, causing a variety of deficits. In order to characterize these deficits, a number of mouse models—such as the 5xFAD strain—have been developed; however, most studies have focused on motor behaviour, cognition, or sensory perception, but little consideration has been given to social deficits. Mice—like humans—are highly social animals, and research has linked social isolation and defeat with a number of behavioural, cognitive, and physiological changes. As a result, this study aims to elucidate the impact of neurodegeneration on social behaviours in the 5xFAD mouse strain. In particular, this work examines responses to olfactory social cues, social vs nonsocial preference, social memory, and aggression in male and female 5xFAD transgenic and wild-type control mice at 3, 6, 9, and 12 months of age.

T3) RadarWatch: Validating an attention training game with a focus on vigilance decrement

Patrick E, Drake R, Kroeker J, Chisholm J, Klein R, Westwood D, Eskes G

Attention deficits can occur with medical, neurological or psychiatric illnesses and limit a person's independence and quality of life. Thus, rehabilitation of these deficits would be beneficial. Posner and colleagues have provided a framework for the attention system that involves three independent but interacting attentional brain networks. The alerting or vigilance network underlies the ability to develop and sustain a state of mental alertness, which consequently produces more rapid detection, selection, and responding. We have recently developed a vigilance game, Radar Watch, that involves tracking a radar screen for designated targets over extended periods of time. Typically in vigilance tasks, the ability to detect and respond to targets decreases as time on the task increases, a phenomenon known as the vigilance decrement. Research has demonstrated that this decrement can occur within the first 5 – 30 minutes of a task and is influenced by how often a target appears. The purpose of the current study was to examine for the presence of a vigilance decrement in Radar Watch. We also investigated if the target rate (high or low) influenced the vigilance decrement. Twenty young adults (mean(SD) age = 20.7(4.3) yrs) were randomly assigned to one of two groups, high target rate or low target rate, and were asked to monitor the radar display for a designated target for approximately one hour. Reaction time (RT) and accuracy of target detection was measured over 6 blocks of 60 radar sweep trials. Preliminary but promising results, as well as future applications, will be discussed in the context of the alerting network and stroke rehabilitation.

T4) God and Karma: Exploring Generosity Levels among Christians and Buddhists in Hong Kong

Curtis R

In most religions around the world, generosity toward others is a moral virtue. Social psychologists are currently working to better understand the factors motivating religious believers to act prosocially toward their ingroup members, as well as others. This study investigated generosity behavior among 40 Christians and 40 Buddhists in Hong Kong, China. Participants played four iterations of a dictator game, in which they divided the equivalent of one day's wage between themselves, a nearby member of their own religion, a faraway member of their own religion, and a faraway member of a different religion. Buddhists kept significantly less money for themselves than Christians. I discuss possible reasons for this outcome, including belief types, religious identity, and methodological concerns. My results highlight a need within the psychology of religion for additional research investigating motivating factors among believers of religions with agentic versus non-agentic moralizing forces.

T5) Item-based Rasch Analysis of the Merrill-Palmer-Revised Scales of Development for Children with Autism Spectrum Disorder

Dempsey EE, Smith IM

Gold-standard diagnostic assessment for children suspected of having autism spectrum disorder (ASD) includes cognitive testing. Psychometrically strong cognitive measures are also indispensable in longitudinal research. Assessment of cognition in ASD is complicated by variable skill levels and uneven profiles of strengths and weaknesses, as well as inherent social communication differences. Few tests have been validated for use with this population. The Merrill-Palmer-Revised Scales of Development (M-P-R) is a standardized cognitive assessment that is well-validated and reliable for use with typically developing preschoolers. However, the suitability of the M-P-R for assessing children with ASD has not been well-researched. The current study's aim is to use an application of Item Response Theory, Rasch Analysis, to investigate the psychometric properties of the M-P-R in a large group of preschoolers with ASD. This presentation will focus on preliminary results of the Rasch Analysis.

T6) When *white* is red in the eye of the *Drosophila* beholder, and philosophy 101

Shaw S

A persistent belief in the fly's-eye community has been that certain useful, easily assayed electrical responses come from particular neurons lying postsynaptic to the fly's photoreceptors -- vertebrate bipolar neuron equivalents. This belief stems from two failures, one observational and one 'philosophical'.

The supposed gold-standard test for a feature like this electrical signature is to delete the putative gene for it (*ort1* in *Drosophila*), show that responses vanish, then reintroduce *ort1* again by clever genetics, rescuing responsiveness. In order to mark reintroduction, groups who did this used a gene marker along with *ort1*, the readily detectable white-eye phenotype. This assumes that the electrical responses from *white* are identical to those from red-eyed, wild-type flies, but the ON-OFF visual responses from the latter reveal very different properties from those in *w¹¹¹⁸* flies: they must have different cellular origins. Despite its cachet, the deletion-rescue formula fails to disclose the origins of visual responses in non-mutant flies.

As to why "gold-standard" genetics produces disappointing results, the formula fits a category well known to logicians -- "the fallacy of affirming the consequent". If you assume that there is only a single solution to the question posed when there actually could be 2 or more alternative solutions, you've bought into your own story, #1. The correct answer could instead be #2 (or #3...), which you failed to eliminate.

T7) FoxP2 Protein Expression in Zebra Finches Reared by Single Parents

Gayda E

The zebra finch (*Taeniopygia guttata*), a sexually dimorphic songbird native to Australia, is one of the best-studied models of vocal learning. Finches have specialized neural circuitry critical for song learning; of significant importance is the basal ganglia component, Area X. In addition, songbird brains express FoxP2, a transcription factor that belongs to a family of genes involved in vocal communication in both humans and songbirds. Levels of FoxP2 protein in Area X decrease across the period of song development, however, it is unclear whether abnormal rearing, such as lack of appropriate tutor modelling, affects typical FoxP2 protein expression. Therefore, the purpose of this study was to analyze FoxP2 protein expression in zebra finches reared in typical biparental nests, and atypical paternal-only, or maternal-only nests. To do so, FoxP2 immunohistochemistry and quantification was completed with birds that had been reared to adulthood under these conditions. FoxP2 protein expression was not robust enough to detect Area X as a discernable nucleus in any birds. Therefore, an exploratory search for FoxP2 protein labelling was completed. There was evidence of lateralized expression of FoxP2 in the striatum of typically reared birds, but birds reared by a single father or single mother did not show this lateralization.

T8) Cue-induced Functional Connectivity in the Nucleus Accumbens and Anterior Cingulate Gyrus in Smokers with and without Psychosis

Ellsworth-Spotton G

The rate of smoking in patients with psychosis is 70 % (Moran, Sampath, Stein & Hong, 2012). Smoking stimuli presented to smokers will elicit a craving response (McBride, Barrett, Kelly, Aw, & Dagher, 2006). The mesolimbic dopamine pathway is implicated in smoking and psychosis and hyperdopaminergia is seen in both groups (Chambers, Krystal & Self, 2001; Fonder et al., 2005). Six patient smokers and six control smokers underwent an fMRI, during which participants were presented with cues. fMRI data was analyzed using FSL and the anterior cingulate gyrus and nucleus accumbens were regions of interest (ROI) for seed based functional connectivity. Patients demonstrate a higher degree of functional connectivity between the ROIs and brain regions involved in emotion, memory and craving. This research broadens the understanding of the neurobiological mechanisms behind the high smoking rate in psychotic individuals and helps to gain a greater understanding of the addiction pathways in these individuals.

T9) Employing Rescorla's truly random condition to measure truly exogenous covert attention

Habibnezhad M, Lawrence M, Klein R

In studies measuring the exogenous attention, cues are employed to capture attention. Ideally, cues are supposed to be both spatially and temporally uninformative of the target stimulus. Studies carried out to measure exogenous attention have successfully made the cues spatially uninformative, while temporal contingency between the cue and the target has been minimized with different measures, none of which guarantees complete uninformative of the cues. The present study used Rescorla's truly random condition to produce such cues.

T10) Pediatric Insomnia Treatment in the Context of Obstructive Sleep Apnea: A Usability and Pilot Study

Isaacs J, Corkum P, Godbout R, Witmans M

1-5% of children suffer from obstructive sleep apnea (OSA), a disorder marked by upper airway blockage during sleep. Over 50% of children with OSA exhibit insomnia symptoms. Of the children with both OSA and insomnia, 20-25% still have insomnia after OSA treatment (i.e., CPAP, adenotonsillectomy). Access to treatment for residual insomnia among children previously treated for OSA is limited. Better Nights, Better Days – Obstructive Sleep Apnea (*BNBD-OSA*), is a parent-directed online intervention for treating residual insomnia following successful OSA treatment. The intervention represents an extension of *BNBD*, an online program which treats primary childhood insomnia. In order to test the usability and initial success of *BNBD-OSA*, 10 parents will complete the program and describe their experiences with it. Changes in children's sleep patterns will be directly measured with a sleep diary and actigraphy. *BNBD-OSA* will allow distance treatment for pediatric insomnia in the context of OSA, significantly reducing the personal and economic costs of childhood sleep disorders.

T11) The influence of Pavlov on Lashley and Hebb

Brown RE

Ivan P. Pavlov was awarded the Nobel Prize for Physiology or Medicine in December, 1904 "in recognition of his work on the physiology of digestion". By this time Pavlov had started to study reflexes and he gave the first definition of the "conditioned reflex" and described how he came to study "psychical reflexes" in his "Madrid speech" (April 1903). In 1906, Pavlov [spelled Pawlow] gave the Huxley Lecture, entitled "The scientific investigation of the psychical faculties or processes in the higher animals" which was published in *The Lancet* (6 Oct. 1906). The English-speaking world thus became aware of Pavlov's work on conditioned reflexes, and in 1909, Robert M. Yerkes and Sergius Morgulis published a review of Pavlov's research in *The Psychological Bulletin* entitled "The method of Pawlow in animal psychology". One of the earliest proponents of classical conditioning in the USA was John B. Watson, the founder of "Behaviorism", whose 1915 Presidential Address to the American Psychological Association was entitled "The place of the conditioned reflex in Psychology" (*The Psychological Review*, March 1916). Watson described his experiments on conditioned motor reflexes using the method of Bechterev, and the research of Karl S. Lashley on conditioned salivary reflexes in humans. Watson conducted his most infamous experiment on "Conditioned emotional reactions" on "little Albert" in 1920. Pavlov visited America in 1923 and 1929 and his Lectures on conditioned reflexes were published in English in 1928. As discussed by Ruiz et al (2003. *Spanish J. Psychol.* 6, 99-111), Pavlov provided American psychologists with three approaches to the study of brain and behaviour: (1) an objective philosophical approach as opposed to subjective methods such as introspection; and (2) a methodological approach through the use of conditioned reflexes. While these were embraced by Americans, (3) the theoretical approach to the study of higher nervous activity through conditioned reflexes, was not so readily embraced. Although Pavlov's methods of conditioned reflexes influenced the theories of Edwin Guthrie, B.F. Skinner, Clark Hull and Kenneth Spence, Guthrie and Lashley (1930) were critical of Pavlov's theory of neural function. Pavlov did not believe in the American theory of Behaviorism and wrote a scathing reply to these critiques (1932).

Meanwhile, Pavlov's student, Boris Babkin moved to Canada where he influenced Donald Hebb. Hebb developed a neurophysiological theory that integrated the ideas of Pavlov and Sherrington on reflexes with the theories of Lashley, Kohler and Tolman into a coherent theory that provided a neural mechanism that explained perception, attention, learning and memory. Hebb considered the neural mechanisms of perceptual integration and came up with a theory that reconciled the two opposing theories of neural function in associative learning: the S-R telephone exchange theory and the field theory of equipotentiality. Hebb developed three ideas to provide a "modern" view of the neural changes underlying cognitive function: (1) the "Hebb synapse" or "Hebb learning rule"; (2) the cell assembly and (3) the phase sequence.

The Rescorla - Wagner model (1972) stimulated new behavioural research into classical conditioning while neurobiologists use classical conditioning to study the neural mechanisms of learning and memory in both invertebrates and vertebrates. Today Pavlovian conditioning is used to study the neurobiology of learning and memory at four levels: (1) behavioural; (2) neural systems and circuits; (3) cellular mechanisms of synaptic plasticity; and (4) molecular mechanisms of neural plasticity underlying learning and memory. Of note are Kandel's studies of the cellular and molecular mechanisms of memory in *Aplysia*; the work of LeDoux, Faneslow and others on the

neural circuits of conditioned fear and the work of Thompson on the role of the cerebellum in conditioned motor responses. Pavlovian conditioning is also used to understand abnormal behaviour and to develop clinical methods of behaviour therapy. Thus, much of the history of Psychology and Neuroscience in America has been influenced by the concepts of classical conditioning as developed by Pavlov, criticized by Lashley and incorporated into Hebb's neurophysiological theory.

T12) The impact of fetal and neonatal experience on mammalian brain development and stress responsivity

Lee K, Weaver I, Goguen D

In humans, experimental and prospective epidemiological research suggests that gestational maternal stress has been shown to produce stable, long-term effects on the physical and psychological development of offspring. Our current research focuses on the role of the transcription factor and chromatin remodeling protein, ATRX, in hippocampal neuron survival in response to psychological stress exposure during the third trimester. Using wild-type and ATRX heterozygous mice, we will examine the effects of psychological gestational stress (PSG) on the quality of maternal care, and on cognitive ability and the expression of anxiety-like behaviors in offspring. Additionally, we will attempt to reverse the effects of PSG by performing rescue experiments either by ATRX overexpression, or by H2AX ubiquitination/histone deacetylase inhibition to promote repair protein recruitment to DNA damage sites in vivo.

T13) Emotion Regulation in Couples Affected by Female Sexual Interest and Arousal Disorder

Dubé J, Rosen NO

Introduction. Female sexual interest/arousal disorder (SIAD) is a sexual dysfunction characterized by absent or significantly reduced sexual interest or arousal, accompanied by significant distress. Women with clinically low sexual desire and arousal have reported more difficulties with emotion regulation than controls; however, whether emotion regulation was associated with women's (and partners') adjustment to the condition has not been assessed. The symptoms of SIAD typically occur in the context of a romantic relationship, meaning that partners are also impacted and will have their own emotional responses to the condition. Therefore, it is essential to take a dyadic approach – which past research has neglected to do – in the study of emotion regulation in SIAD.

Aim. The current study aims to investigate how emotion regulation in couples affected by SIAD relates to their relational and sexual wellbeing (e.g., sexual satisfaction, distress, and desire).

Methods. Women with SIAD and their partners ($N=102$) will complete standardized self-report questionnaires assessing emotion regulation, relationship satisfaction, sexual satisfaction, sexual desire, and sexual distress. Data will be analyzed per the Actor-Partner Interdependence model using multilevel modeling.

Predicted Results. Individuals with more effective emotion regulation will report greater relationship satisfaction and sexual wellbeing. Further, individuals with more effective emotion regulation will have *partners* who report greater relational and sexual wellbeing, compared to those with poorer emotion regulation.

T14) The Moderating Role of Self-Compassion on the Relation between Alcohol-Involved Sexual Assault and Negative Emotional Outcomes in Undergraduate Men and Women

Strickland NJ, Stewart SH, Kehayes I-L, Wekerle C, Moore C, Caring Campus Team

Approximately 15% of students have experienced alcohol-involved sexual assault (AISA; Elliott et al., 2004). Victim-drinking AISA may result in higher rates of self-blame, and self-compassion (SC; i.e., extending the same kindness and care towards oneself that one would to someone else in a similar situation) may promote resilience by attenuating the association between AISA and negative emotional outcomes. Undergraduates ($N = 1,255$) completed measures tapping past-term victim-drinking AISA (see Kehayes et al., under review), SC (i.e., Self-Compassion Scale; SCS; Neff, 2003b), anxiety and depression (K-10; Kessler et al., 2002), and self-perceived mental health (Ahmad et al., 2014). It was hypothesized that: AISA would be positively associated with anxiety, depression, and poor self-perceived mental health (H1), high SC would be positively related to mental health, and negatively related to anxiety and depression (H2), and SC would moderate (i.e., buffer) the association between AISA and these negative outcomes

(H3). Preliminary results showed that, partially supportive of H1, AISA was related to higher depression ($r = .07, p < .05$). In addition, consistent with H2, high SC was associated with better mental health ($r = .56, p < .05$), and lower depression ($r = -.61, p < .05$) and anxiety ($r = -.47, p < .05$). In partial support of H3, tests of moderation showed a significant interaction between the self-criticism scale of the SCS and AISA on self-perceived mental health. However, instead of the hypothesized buffering effect, it was found that endorsing the self-criticism subscale of the SCS predicted lower levels of mental health, regardless of whether AISA was experienced, but those who had low levels of self-criticism only had poor mental health if they have experienced AISA. These results suggest that for people who are not already self-critical, experiencing AISA can lead to worse outcomes. Results will be further explored following confirmatory factor analyses of the SCS.

T15) Behavioural, Molecular and Epigenetic consequences of Perinatal Infection throughout Development and into Adulthood.

Wheeler RV, Alam S, Weaver ICG, Franklin TB

Human studies have indicated a link between perinatal infection (PI) and the development of autism-spectrum disorder (ASD) suggesting an epigenetic etiology. Similarly, mice exposed to PI have an ASD-like phenotype, behaviourally and molecularly. In the current study, we injected pregnant C57/BL6 mice with lipopolysaccharide or polyinosinic:polycytidylic acid to induce a bacterial or viral response, respectively. The offspring were tested during development and at adulthood for social communication and interaction, and repetitive behaviours. Additionally, the expression of several synaptic and regulatory epigenetic genes were analyzed in the brains of offspring at post-natal day zero. PI mice displayed abnormal social behaviours, repetitive behaviours and gene expression compared to control mice. Future directions investigating the potential for PI-induced epigenetic alterations will also be discussed.

T16) Is there a relationship between working memory and reading in university students?

Rehan S, Rubinfeld L, Eskes G

University students with a history of reading difficulties (HRD) experience reading and academic difficulties in the university setting and are in need of support; however, without a diagnosis of a learning disability, they are not eligible for academic accommodations. A critical relationship between working memory and reading has been well-documented and working memory (WM) deficits are common in individuals with reading deficits. We are conducting a study that aims to characterize the reading, attention, WM abilities of HRD undergraduate students and to investigate the effectiveness of core WM training in improving these students' working memory and reading abilities. As a first step, we have explored the relationship between WM and reading in a control group of 23 undergraduate students without a history of reading difficulties (NRD; M(SD) age = 20.7(1.1)). Hierarchical linear regression analyses revealed that verbal working memory performance using the Operation Span was a significant predictor of performance on all measures of word reading ($\beta = .49$), non-word reading ($\beta = .59$), reading rate ($\beta = .61$), and reading comprehension ($\beta = .76$). In contrast, visual working memory performance on the Symmetry Span was not a significant predictor of reading performance ($ps > .05$). Results confirm the relationship between verbal working memory and reading and support our plan to investigating WM training in HRD undergraduate students.

T17) Male Experience, Female Investment and Offspring Anxiety Behavior: Where Father's Nature Meets Mother's Nurture

Korgan A

Detecting past experiences may inform a female about prevailing ecological conditions, in addition to paternally-induced phenotypes that may be disseminated to offspring. We examined whether a male rat's exposure to metabolic stressor (high-fat diet; HFD) influences a female's attraction towards a male, subsequent mother-infant interactions and the development of defensive (emotional) responses in the offspring. Females displayed less interest in males following HFD conditions. Mothers that reared young in larger, semi-naturalistic housing provided more licking and grooming and active arched-back-nursing behavior toward their offspring compared to dams housed in standard

housing, although some effects interacted with paternal diet exposure. Paternal diet and maternal rearing environment revealed sex-dependent differences in offspring weight, juvenile social interactions and anxiety-like behavior in adolescence. Additionally, paternal predator experience and maternal housing independently affected variations in *crf* gene promoter acetylation and *crf* gene expression in response to an acute stressor in offspring. Our results show for the first time in mammals that male HFD consumption may contribute to stable behavioral variation among females in preference for mates and maternal care, even when the females are not directly exposed to HFD. These results, together with our previous findings, suggest that paternal HFD exposure and maternal rearing conditions can influence maternal behavior and development of defensive responses of offspring.

T18) Perfectionistic Concerns Confer Risk for Anxiety Symptoms: A Meta-Analysis of 11 Longitudinal Studies

Vidovic V, Smith MM, Sherry SB, Stewart SH, Saklofske DH

Over 30 years of case reports, theoretical accounts, and empirical research implicate perfectionism in anxiety, yet the extent to which perfectionism predicts longitudinal changes in anxiety remains unclear. To address this, we conducted a meta-analysis of research testing whether socially prescribed perfectionism, concern over mistakes, doubts about actions, self-oriented perfectionism, and personal standards predict change in anxiety, after controlling for baseline anxiety. Our literature search yielded 11 relevant studies for inclusion ($N = 3,144$). Meta-analysis, using random-effects models, revealed socially prescribed perfectionism, concern over mistakes, and doubts about actions, but not self-oriented perfectionism or personal standards, displayed small positive relationships with follow-up anxiety, after controlling for baseline anxiety. Thus, our study highlights the unique role of different perfectionism dimensions in psychopathology, and offers novel insight into the pre-morbid personality of individuals particularly prone to symptoms of anxiety.

T19) Examining Links Between Sexual Savouring and Sexual and Relationship Well-Being

Richardson HM, Vannier SA, Rosen NO

Savouring refers to an individual's active ability to anticipate, focus on, and appreciate positive experiences. Savouring has three components: anticipation (e.g., enjoyment of events before they occur), savouring the moment (e.g., prolonging enjoyment by own effort), and reminiscing (e.g., enjoying looking back on happy times). Savouring has been linked to positive outcomes such as greater life satisfaction, self-esteem, and happiness. There has been no research examining savouring of positive sexual experiences, however, the previous literature linking savouring to general positive outcomes suggests that more savouring may promote greater sexual and relationship well-being. In the current study, 303 participants completed online measures assessing sexual savouring, distress, frequency, functioning, satisfaction, and relationship quality. Sexual anticipation was associated with higher sexual functioning and satisfaction. Sexual savouring of the moment was associated with higher sexual frequency, functioning, and satisfaction, greater relationship quality, and lower sexual distress. Sexual reminiscing was associated with lower male sexual functioning. These results suggest that individuals who anticipate and savour their positive sexual experiences may experience greater overall sexual and relationship well-being.

T20) Self-Distracton During Exercise in Clients with High (versus Low) Anxiety Sensitivity

Pridy C, Watt M, Stewart SH

Anxiety sensitivity (AS) is fear of sensations experienced during emotional arousal or physical activity. High AS is associated with increased risk of psychopathology (Reiss, 1991). People with high AS avoid exercise because it induces feared sensations (Smits, et al., 2010). Running has, however, been employed as interoceptive exposure (IE) to enhance efficacy of a brief CBT intervention for high AS (e.g., Watt et al., 2006). Self-distracton during IE trials may enhance or detract from their effectiveness. Stevinson and Biddle (1998; 1999) outlined a comprehensive model of four exercise-related cognitive styles combining the two dimensions of attentional focus (internal/external) and task relevancy (relevant/irrelevant). The proposed project will develop and validate a 20-item Attention and Distraction during Exercise Questionnaire (ADEQ), tapping the four cognitive styles outlined by Stevinson and Biddle, plus

qualities of music used for listening during exercise. We will test predictions that high (vs. low) AS participants will favour a self-distractive, externally-focused task irrelevant cognitive style; that high AS participants will prefer an externally-focused and task irrelevant style when deliberately self-distracting from distressing interoceptive cues or an internally-focused and task-relevant style when catastrophizing about feared exercise-induced sensations; and that an externally-focused and task irrelevant style will moderate the known, direct relationship between AS and exercise avoidance. Baseline data gathered during the proposed study will assist in future efforts to determine whether limiting self-distraction during an exercise-based intervention for high AS can help to maximize treatment benefits.

T21) Investigating the neural signature of Inhibition of Return using a multimodal paradigm

MacNeil H, Klein R, d'Entremont G

Inhibition of Return (IOR) is the phenomenon whereby attention is inhibited at locations that have previously been attended to as a mechanism to reduce unnecessary attention. Previous research looking at event-related potentials (ERPs) using electroencephalography (EEG) has suggested that P1 reductions are the electrophysiological markers of IOR. However, recent work has demonstrated that P1 reductions may simply be the result of repetitive sensory stimulation rather than of IOR itself. This suggestion has been examined using multi-modal IOR paradigms to compare P1 components in conditions in which the modality is repeated to conditions in which IOR is generated across modalities. The present study aimed to expand on these findings. It was hypothesized that IOR would be observed across all four conditions of Cue Modality and Cue-Target Onset Asynchronies (CTOA) of either 700 ms or 1000 ms, and that P1 reductions would only be seen in the visual-visual conditions. The results supported the hypothesis that IOR effects would be generated across all four conditions. The effect of Cue Modality on P1 amplitude was insignificant, however it did show a trend suggesting increased P1 cueing effects for the Visual Cues compared to the Tactile Cues. The results of this study lay out the framework to further test this hypothesis in future research using a larger sample size.

T22) On the choice of animal models in neuroscience

Mitchell DE

I recently was a member of a panel that reviewed the value of various animal models for study of the neural basis of human amblyopia and its treatment. This talk will summarize our findings as well as our recommendations for future research to provide cohesion across animal models and suggestions for a protocol to guide the steps towards translations of results from animal models to human clinical trials. Our recommendations on the latter may generalize to animal models of other neurological disorders.

T23) The roles of parvalbumin and somatostatin expressing interneurons in modulating contrast adaptation in mouse primary visual cortex

King JL, Korgan AC, Crowder NA

Contrast adaptation is a visual phenomenon induced by prolonged viewing of a high contrast stimulus, which then decreases the perceived contrast of a subsequently presented stimulus. At the cellular level it has been well established that contrast adaptation causes characteristic shifts in the contrast response functions of primary visual cortex (V1) neurons, but the neural circuitry responsible for this shift is still unclear. Our lab recently demonstrated that when V1 is silenced during adaptation, via optogenetically stimulating GABAergic interneurons, the magnitude of adaptation is attenuated, and the adaptation that does occur is similar to that measured in thalamic inputs. The current work is aimed at building on these findings and untangling the local interneuron circuitry contributing to contrast adaptation in V1. To determine the roles of specific GABAergic interneurons, we partially suppressed V1 during the adaptation period by weakly activating either parvalbumin (Pvalb+) or somatostatin (Sst+) expressing interneurons optogenetically. Our preliminary results suggest that neither Pvalb+ or Sst+ neurons are solely responsible for the shift in neural contrast response functions after adaptation, but the gain control mediated by Pvalb+ interneuron activation has a larger impact on adaptation than Sst+ activation.

T24) The Effect of French Immersion on Native English Processing in the Brain

Tagarelli KM, Lazar-Kurz Z, Newman AJ

French Immersion programs have been in place across Canada since the 1970s, and they are particularly popular in the Maritimes, where up to 32% of English-language schools have such programs. Beyond developing second language skills, students in French Immersion programs have been shown to outperform their English-only peers in first language and other academic skills. However, we know virtually nothing about how this early and prolonged exposure to a second language influences brain development and processing. The current study uses event-related potentials to examine language processing in individuals who have gone through French Immersion, compared to those who have not.

T25) Exploring Parent-Reported Perioperative Distress Behaviours in Children with Autism Spectrum Disorder

Cameron E, Snow S, Smith I, Chorney J

The perioperative experience can be distressing for children. Reports state that between 40% and 50% of typically developing children experience perioperative distress. While perioperative distress has been well outlined in typically developing children, similar observations have not been made in children with autism spectrum disorder (ASD; a heterogeneous neurodevelopmental disorder).

This study aimed to report the frequency of, and define distress behaviours in children with ASD. Participants in this study included 45 families of children with ASD undergoing day surgery in the IWK Health Centre. Parents completed the Perioperative-Distress Questionnaire to assess their child's perioperative distress.

Ninety percent of children with ASD experienced distress during one-or-more time points throughout the surgical experience. Commonly reported distress behaviours included: disorientation, aggression, and verbal requests to leave. This is the first study to examine the frequency of, and describe perioperative distress in children with ASD. Children with ASD seem to have greater perioperative distress levels compared to typically developing children.

T26) Potential molecular conflict between detoxification and immune defense in *Manduca sexta*

Hall LE, Adamo SA

Like other animals, larval *M. sexta* lose their appetite after an immune challenge, yet the biological function of this illness-induced anorexia is poorly understood. This research examines the hypothesis that illness-induced anorexia allows the caterpillar *M. sexta* to shift molecular resources from food detoxification to immune function. Caterpillars that received an immune challenge were more likely to die after ingesting permethrin than those that did not. Conversely, a sublethal dose of permethrin reduced resistance to the Gram negative bacterium *S. marcescens*. These results suggest that there is a negative interaction between detoxification and immune defense. We used qPCR to test for a potential molecular conflict between detoxification and immune pathways. While expression of immune genes appear unaltered by a combined immune-toxin challenge, expression of certain detoxification genes are lessened in the presence of a combined challenge. These results indicate a potential molecular resource trade-off between immunity and detoxification pathways.

T27) Illness-induced anorexia may help promote immune function by reducing the use of glutathione, a limited resource, by detoxification pathways.

Miller D, Hall L, Adamo S

In response to an immune challenge, most animals have reduced appetite, a behaviour called 'illness induced anorexia'. Despite the ubiquity of this response, its potential biological function is not well understood. The caterpillar *M. sexta* shows increased susceptibility to pesticides when the immune system becomes activated. Similarly, sublethal amounts of pesticide increase susceptibility to disease. Recent qPCR data suggests a molecular

conflict between the pathways for food detoxification and immune defense, centered on the use and availability of glutathione (GSH). GSH is an antioxidant tripeptide used in both detoxification and immune-response pathways. To test the hypothesis that limitations in the availability of GSH leads to trade-offs between immune function and detoxification, we compared whether supplemental GSH could reduce the increased morbidity and mortality of caterpillars given both an immune and toxic challenge simultaneously. Preliminary data suggest that supplemental GSH can rescue caterpillars exposed to both challenges, suggesting that GSH may drive a tradeoff between the two pathways. Illness-induced anorexia would shift GSH resources toward immune function.

T28) Susceptibility to monocular deprivation following immersion in darkness just prior to the critical period peak

Lingley A, Duffy KR

Immersion of animals in complete darkness reveals that the maturation and plastic capacity of visual circuitry depends on binocular visual experience. Recent investigations have shown that imposing 10-days of darkness immersion during juvenile life instates a neurobiological context wherein an animal's susceptibility to subsequent modification of visual experience is augmented. For instance, our previous study showed that immersion of 12-week old cats in complete darkness for 10-days promotes neurobiological changes that render the effects of subsequent monocular deprivation (MD) much more severe than those observed in age-matched controls. This result suggested that darkness immersion promotes an animal's capacity for plasticity, recapitulating a developmental state akin to a much younger animal. The current study sought to explore whether short-term darkness immersion could similarly promote visual plasticity at the peak of the critical period (p30), when an animal's sensitivity to visual perturbation is at its biological maximum. Although 7-days of MD at this age produced potent alterations of neuronal soma size in the dorsal lateral geniculate nucleus (dLGN), 10-days of preceding darkness immersion had no influence on the subsequent MD effect. Similarly, MD imposed following darkness-induced changes in dLGN neurofilament immunoreactivity equivalent, but not greater than those observed in animals subject to MD alone. These results suggest that the propensity of darkness to promote susceptibility to MD depends on the animal's initial capacity for visual plasticity. Further, it appears that the ceiling for critical period plasticity may be constrained by factors resistant to the effects of darkness.

T29) Drinking frequency, drinking quantity, and alcohol-related problems as trait-states: A 4-wave, 18-month longitudinal study of female undergraduates

Bartel, SJ, Sherry SB, Nealis LJ, Collins JL, Lee-Baggley DL, Stewart SH

Background: Drinking behavior and alcohol-related problems in female undergraduates are topics of growing interest and importance. Research in this area widely assumes that alcohol frequency, alcohol quantity, and alcohol-related problems are trait-like (i.e., strongly temporally stable). Under the assumption of trait-like stability, a single assessment is enough to capture a person's usual pattern of drinking behavior and alcohol-related problems; however, in making this assumption, research in this area fails to consider that there may be substantive state-like variability in drinking behavior and alcohol-related problems. To address this shortcoming, we tested a trait-state model of drinking behaviors and alcohol-related problems in female undergraduates using a 4-wave, 18-month longitudinal design. We hypothesized alcohol frequency, alcohol quantity, and alcohol-related problems would show both trait-like stability and state-like variability across our four waves of data collection, thereby supporting the trait-state model. Methods: A sample of 207 female undergraduate drinkers completed self-report measures of alcohol frequency, alcohol quantity, and alcohol-related problems four times (i.e. every 6-months for 18 months). Results: Generalizability theory analyses indicated alcohol frequency, alcohol quantity, and alcohol-related problems all exhibited both trait-like stability and state-like variability. While a greater proportion of the variance in alcohol quantity and alcohol-related problems was attributable to trait-like stability, a greater proportion of the variance in alcohol frequency was attributable to state-like fluctuations. Conclusions: Drinking patterns of female undergraduates tend to fluctuate over time, particularly frequency of use. It should not be assumed that a single assessment of drinking behavior and alcohol-related problems is a valid representation of a young woman's drinking patterns.

Keywords: trait, state, trait-state, alcohol, alcohol problems, generalizability theory

T30) Orthographic knowledge as a predictor of gains in lexical access

MacKay E, Deacon H, Conrad N

Word reading is a foundational skill in children's reading development. One construct related to and important for word reading is lexical access. Lexical access is defined as the activation and/or retrieval of words in the mental lexicon (Perfetti, 1992). Theoretically, orthographic knowledge, or the knowledge of specific and general letter patterns, has been suggested as important for successful and fast lexical access. As such, we will longitudinally examine if orthographic knowledge predicts gains in fast lexical access for 104 English-speaking children from grades four to five. We will do so with an autoregressor cross-lag design. To form our composite of orthographic knowledge, we administered measures of lexical and sublexical orthographic knowledge in grade four. To assess lexical access, we administered a measure of word naming measure to students in grade four and five. A measure of phonological awareness and a measure estimating nonverbal ability were administered as controls. To examine if orthographic knowledge predicts gain in fast lexical access, we will employ Structural Equation Modeling. Our model will include orthographic knowledge at grade four as the independent variable and lexical access at grade five as the dependent variable. Fourth grade lexical access, phonological awareness, and nonverbal ability will be entered as controls. Given theoretical predictions, we expect to find that orthographic knowledge predicts gains in speed of lexical access. The results of this study will inform key theories of reading development by providing novel support for the role of orthographic knowledge in successful and timely lexical access.

T31) The Connectome of a *Ciona intestinalis* larva

Ryan K

The nervous systems that drive animal behaviour are founded on synaptic connections between cells therein. Small nervous systems allow us to examine the entirety of a nervous system in great detail, revealing both ultrastructural and network details. We have used anatomical connectomic approaches by means of serial-section EM (ssEM) to analyze the number and distribution of synapses within the networks of the larval brain of *Ciona intestinalis*. Larval responses to environmental cues rely on the network of underlying neuronal connections that translates these cues into motion. The anatomical connectome reveals the components and connections of pathways for sensory integration of visual, gravity, and peripheral information both to and from pathways of other sensory systems. These sensory pathways feed into motor networks described through the current analysis that are involved in central pattern generation and a putative escape response. Cell by cell analysis of morphology and all synaptic connections enables the identification of new cell types and the classification of all neurons in this small model chordate nervous system. The network complexity allowing integration at multiple levels provides a substrate for more complex larval behaviour based on interactions between pathways that converge on the motor generating network of the *Ciona intestinalis* larva.

T32) A second complete connectome: the larval CNS of the ascidian *Ciona intestinalis*

Meinertzhagen IA, Lu Z, Ryan K

The diminutive brains of invertebrates, especially those of larval stages with few cells, are Nature's bounty for connectomic analyses. Pride of place for the first complete connectome report goes to *C. elegans* (White et al., 1986, *Phil. Trans. Roy. Soc.*). *Drosophila* also supports powerful genetic methods and interesting behaviours, especially for vision, olfaction and courtship, but its neurons branch profusely to yield many slender neurites that require special methods to reconstruct at EM level. Pursuing an alternative possibility of an interesting simple brain in a model species, we have used ssEM to compile a complete densely reconstructed connectome for the CNS in the tadpole larva of *Ciona intestinalis* (Ryan et al., 2016, *eLife*), a chordate tunicate, a sister group of vertebrates. The dorsal, tubular CNS of the ~1-mm tadpole larva is built on a similar plan to vertebrate brains, its ~330 cells distributed rostrocaudally in three centres: a brain vesicle, motor ganglion and caudal nerve cord. Although early neural development is well studied, detailed cellular organization of the swimming larva's CNS is hitherto unreported. Neurons are unipolar with at most a single dendrite and few synapses. The 177 neurons formed 6618 CNS synapses including 1772 neuromuscular junctions, augmented by 1206 gap junctions. Some synapses are

bidirectional, others form reciprocal or serial motifs; 922 were polyadic. Axo-axonal synapses predominate. Most neurons have ciliary organelles, and many features lack structural specialization. Glia and myelin especially are both lacking. Despite equal cell numbers on both sides, neuron identities and pathways differ left/right, asymmetries probably associated with the extreme reduction in cell numbers. Brain vesicle asymmetries include a right ocellus and left coronet cells. Some neuron types show homology with vertebrate neurons, as revealed by their similar circuits.

Posters

P1) Investigation of neuron structural modifications within the visually deprived superior colliculus.

NESC3370: Zajdlik K, Lausch S, Crouse C, Elia T, Aronitz E, Ellis S, Fish J, Lagman P, MacDonald J, McGroarty E, Mustafa D, Piercey M, Bassil S, Connell L, Lee S, MacLellan K, Pappin J, Ribeiro G, Kraeutner S, Duffy KR

Neurons within the mammalian primary visual pathway exhibit a high capacity for neural plasticity early in postnatal development. Even a short duration of monocular deprivation (MD) can lead to structural and functional neural modifications that result in a profound visual impairment called amblyopia. Although modification of neurons within the superior colliculus (SC) has been suggested to underlie ocular motor deficits associated with amblyogenic rearing, identification of structural abnormalities within the SC has been elusive. In this study we imposed a complete form of MD by inactivating one eye for 10 days. We reasoned that this extreme form of amblyogenic deprivation would be more likely to precipitate measurable structural abnormalities within the SC. Following 10 days of MD we observed a reduction in the size of neuron somata within the contralateral SC that receives most input from the inactivated eye. We also showed that this form of MD is sufficient to cause a loss of neurofilament protein within the SC contralateral to the inactivated eye. These results demonstrate that a profound monocular deprivation is sufficient to catalyze neuron structural modification within the superior colliculus, and this alteration may be enabled by a reduction of neurofilament protein that destabilizes the cytoskeleton and raises the capacity for plasticity.

P2) A Brain Museum Tour of Europe

Brown RE

Europe has a rich history of neuroscience research and clinical neurology, but where can the history of European neuroscience be found? The historical artifacts, documents and discoveries of European neuroscience exist in many museums, but these are often forgotten or neglected within Europe and relatively unknown outside of Europe. The purpose of this project is to present a tour of the brain museums of Europe on a WEBSITE, showing the museums with materials relevant to the history of neuroscience in each country. The history of neuroscience relies on objects from the past and this website describes the collections related to brain research in European museums. Using this website will enable students and researchers to locate historical objects in museums and plan visits to these museums for teaching and research. The presentation will consist of a poster/oral presentation and a website which meeting participants can browse for information. The present poster/Website contains information on 31 brain museums in 18 countries, with more being added as we find them. The website is a work in progress and we hope that users will provide us with information about brain museums which we have not yet discovered. If you are planning a trip to one of the European cities with a brain museum, this website will guide you to the location and the exhibitions on view. Enjoy your tour of Brain Museums in Europe! This project is sponsored by the FENS History of Neuroscience Committee. If you know of brain museums not presented on this poster, please contact Richard Brown at rebrown@dal.ca.

P3) Does active sharing lead to increased positive affect in children?

Puddicombe E, Pratt T, Longard J, Moore C

Although accumulating evidence suggests sharing can lead to increased positive affect in adults and children, it remains unclear whether children must be active participants in the sharing, or if they might show the same affective benefits when observing sharing. The current study investigated whether active sharing has an additional affective benefit above and beyond passively observing sharing or passively receiving rewards. Participants included 63 typically-developing 5- and 6-year-olds. We used a between-subjects yoked design to match participants across the three experimental conditions to control the number of stickers children received across 10 trials of a resource-allocation task. Two independent coders rated children's videotaped facial affect. We found children in the active sharing condition showed significantly higher positive affect than those in the passive sharing condition, and children in the passive sharing condition showed significantly higher positive affect than those in the passive receiving condition. Interestingly, children did not display more positive affect on sharing as opposed to no sharing trials. Another factor, such as active choice, may account for the affective benefit in this context.

P4) eHealth intervention recruitment: Relative success of digital, media, and traditional recruitment methods for Better Nights, Better Days

Rigney G, Chambers CT, Godbout R, Gruber R, Hall W, Reid G, Stremler R, Weiss SK, Witmans M, Begum EA, Corkum P

Introduction

Approximately 25% of children experience insomnia, which can impact their physical and psychosocial health, as well as their caregiver's well-being. Few children receive evidence-based interventions, due in part to a shortage of available resources. Internet service delivery (eHealth) has the potential to be a powerful tool to overcome barriers to treatment. To this end, we developed Better Nights, Better Days (BNBD) - an innovative eHealth program for primary caregivers aiming to provide accessible and evidence-based care for insomnia in typically developing children aged 1-10 years. We were interested in the uptake for this intervention delivery method.

Methods

A randomized controlled trial (RCT) is being conducted to evaluate the efficacy of BNBD, targeting 400 English speaking primary caregivers across Canada. As BNBD is an eHealth intervention, the recruitment strategy was primarily based on online tools. A comprehensive social media campaign was developed and executed in collaboration with social media experts. As part of the online screening questionnaire, individuals were asked to indicate how they heard about the study (e.g., advertisement, Facebook, family member or friend). The success of recruitment was measured based on participant engagement and report.

Results

The BNBD RCT launched September 2016. Within three months over 2,000 individuals entered the study database to begin screening, and the target number of 400 English speaking primary caregivers had consented, with their study participation currently ongoing. Results from the online screening questionnaire indicated 36% heard about BNBD through social media platforms (i.e., Facebook and Twitter), 26% through media and advertisements, and 25% from a family member or friend.

Conclusion

The high rate of recruitment speaks to both the need for intervention, and the effectiveness of online recruitment methods. Connecting to millennial parents requires the use of online resources, particularly social media. This strategy increases study visibility through a dynamic online presence.

P5) On the Time-Course of Cued Temporal Attention

McCormick C, Redden R, Lawrence M, Klein R

A fruitful tool for studying endogenous control of temporal attention is the Kingstone paradigm (1992). Kingstone's cues had two functions: providing information about the foreperiod and signalling to begin timing of that foreperiod. Interested in how exogenous alerting might interact with endogenous alerting we implemented two novel manipulations. First, we separated these two cue functions. Second, borrowing from Lawrence & Klein (2013) we manipulated warning signal intensity: one signal entailed an increase in intensity (I^+), which generates alerting exogenously; the other signal entailed a qualitative change with no change in intensity (NO), which would not be expected to generate alerting exogenously. When intensity was manipulated randomly from trial-to-trial, endogenous shifts of attention in time were unexpectedly absent. We hypothesized that participants invested substantial cognitive resources to detect the NO intensity signals and that interfered with their endogenous utilization of the pre-cues. To test this idea, in the second experiment we blocked the manipulation of signal-intensity. Consistent with our hypothesis, the Kingstone pattern was replicated in the I^+ condition.

P6) Novice Inter-Rater Reliability in Spike Sorting: Tetrodes vs. Single Channel Electrodes

Shapiro JT, Korgan AC, King JL, Crowder NA

Measuring the response properties of individual neurons within the brain remains a fundamental technique in neuroscience. A challenge in analyzing this *in vivo* electrophysiological data, called spike sorting, involves distinguishing which action potentials arise from the jumbled activity of several simultaneously recorded neurons. Most modern spike sorting software uses semi-automated tools that require a skilled analyst, and thus spike sorting accuracy depends on subjective judgments. Multi-channel electrodes (e.g. tetrodes) have the capacity to yield more data than single-channel electrodes, but it may be more difficult for analysts to learn to spike sort multi-channel data because of the additional complexity. In this study two novice analysts with similar levels of training sorted identical recordings from mouse visual cortex that were acquired with single-channel electrodes or tetrodes to determine whether their accuracy was dependent on the type of data being analyzed. The novice analysts performed similarly with both kinds of electrophysiological data

P7) Assessing Attention: A tablet-based proof of concept with the Dalhousie Computerized Attention Battery (DalCAB)

Patrick E, Cutmore L, Kroeker J, Jones S, Salmon J, Eskes G

Attention deficits can occur with medical, neurological or psychiatric illnesses and limit a person's independence and quality of life. We developed the Dalhousie Computerized Attention Battery (DalCAB) to measure various aspects of attention and have established the validity and reliability of a desktop version of DalCAB in healthy younger and older adults. In this project we sought to investigate whether performance on a touch-screen tablet version of the DalCAB would generate similar performance patterns compared to a normative data set collected on a desktop computer. One hundred healthy young adults (mean(SD) age = 22(3.7) yrs) participated in two studies. Study one (n=40) examined the optimal position of the 2-response touch buttons on a Surface Pro 4 touchpad screen and Study two (n=60) explored the effect of tablet position and screen size. Performance patterns were analyzed within each study and also compared to normative data from fifty young adults (mean(SD) age = 21.78(2.9) yrs) previously collected on an Apple iMac desktop version with a mouse. Within study analyses of tablet data showed that all expected performance patterns were obtained consistently across both studies, and these patterns were not strongly affected by the location of the response touch buttons, the tablet placement or the screen size. Reaction time (RT) performance patterns from the tablet were similar to the desktop patterns, although RTs on the tablet were slower overall, sometimes resulting in greater differences between task variables. These results indicate the robustness of attention performance patterns, and the DalCAB tablet version is a promising alternative to the desktop version. Thus, this study extends the potential applicability and marketability of the DalCAB to different populations and environments.

P8) 15 Years of ANT Research: Exploring the evolution of the Attention Network Test

Arora S, Klein R

The Attention Network Test (ANT) is a computer-based tool widely used to disassociate and measure three networks of attention: alerting, orienting and executive functioning. Originally developed 15 years ago by Posner and his colleagues (Fan et al., 2002) the ANT has been cited for use in over 500 experiments. The present survey investigates the evolution of the ANT and the ways in which it has been employed and extended. Through the development of a comprehensive database, we are able to explore: the various modifications from original ANT that have been simulated, the applications to which it has been used, and attention network scores across various conditions, groups, and regions.

P9) Utilization of Research Electronic Data Capture (REDCap) in a web-based Randomized Controlled Trial

Begum EA, Corkum P

Research Electronic Data Capture (REDCap) is a web-based electronic data capture software that allows researchers to build project-specific data collection instruments to manage research data, store data securely, and facilitate data analysis. Its randomization module makes it an ideal tool for longitudinal studies and randomized controlled trials (RCTs). This software is currently being utilized in a national RCT, *Better Nights, Better Days*, coordinated through Corkum LABS. This resource has demonstrated capabilities to facilitate study procedures while maintaining the security of study data, reducing staff time, and incidental costs. Measures are housed in the REDCap platform and administered automatically to participants in a sequential manner throughout the study. Randomization was performed within REDCap. REDCap also provides features to export data in multiple formats (SPSS, Excel, R Statistical Software, SAS, or STATA). The REDCap database can be easily modified for future studies.

P10) Major connections in glomerulus VA1v in the *Drosophila* antennal lobe

Horne, JA, McLin S, Wiederman M, Borycz JA, Tarnogorska D, FlyEM Project Team, Plaza S, Xu S, Hess H, Meinertzhagen IA

The connectome for glomerulus VA1v of the *Drosophila* antennal lobe has been completed. This *fruitless*-positive, sexually dimorphic glomerulus is responsive to fly odour (Sakurai et al. Nat Commun, 4:1825, 2013). 85% of its synaptic connections have been assigned to the three major classes of neurons: olfactory receptor neurons (ORNs), projection neurons (PNs) and local interneurons (LNs); ORNs are predominantly presynaptic and PNs predominantly postsynaptic (Rybak et al. 2016; Tobin et al. 2017). The synaptic organization of ORNs and PNs has been reported for other glomeruli from ssEM, but not for specific PN types, while the anatomical connections of LNs have not so far been reported at all. The ratio of pre- to post connections for ORN's was 4.2, while for PN's was 0.23. LNs had nearly equal numbers of pre- and postsynaptic contacts with a ratio of 0.89. More generally the output from ORNs dominate the connectome with 56.9% of all connections, 37% to the PN's, 14.6% to LNs.